FACULTY OF ENGINEERING
CHULALONGKORN UNIVERSITY
2110201 Computer Engineering Mathematics
Year II, Second Semester, Midterm Examination, Mar 11, 2021 08:30-11:30

Nae.....Student ID..... CR58 .....

Remarks

1. **There are 17 questions in this exam paper, with the total number of 10 pages including this page.**
2. **The exam duration is 180 minutes.**
3. Write your Student ID, full name in the space provided on the top of every page.
4. No documents or calculators are allowed inside the exam room except for non-programmable and non-graphic calculators.
5. Borrowing is not allowed unless it is supervised by the proctor.
6. You must not bring any part of this exam paper outside. The exam paper is a government's property. Violators will be prosecuted under a criminal court.
7. Student who wishes to leave the exam room before the end of the exam period, must raise his/her hand and ask for permission before leaving the room. Student must leave the room in the orderly manner.
8. Once the time is expired, student must stop writing and must remain seated quietly until the proctors collect all the exam papers or given exam booklets. Only then, the students will be allowed to leave the room in the orderly manner.
9. Any student who does not obey the regulations listed above will receive punishment under the Faculty of Engineering Official Announcement on January 6, 2003 regarding the exam regulations.
   a) **With implicit evidence or showing intention for cheating, student will receive an F in that subject and will receive an academic suspension for 1 semester.**
   b) **With explicit evidence for cheating, student will receive an F in that subject and will receive an academic suspension for 1 year.**

ห้ามนิสิตพกโทรศัพท์และอุปกรณ์สื่อสารไว้กับตัวระหว่างสอบ หากตรวจพบจะถือว่านิสิตกระทำผิดเกี่ยวกับการสอบ อาจต้องพ้นสภาพการเป็นนิสิต หรือ ให้ได้รับ F และอาจพิจารณาให้ถอนรายวิชาอื่นทั้งหมดที่ลงทะเบียนไว้ในภาคการศึกษานี้

*** ร่วมรณรงค์การไม่กระทำผิดและไม่ทุจริตการสอบที่คณะวิศวกรรมศาสตร์ ***

ข้าพเจ้ายอมรับในข้อกำหนดที่กล่าวมานี้ ข้าพเจ้าเป็นผู้ทำข้อสอบนี้ด้วยตนเองโดยมิได้รับการช่วยเหลือ หรือให้ความช่วยเหลือ ในการทำข้อสอบนี้

1. Pencil is allowed.
2. Write your answer in the provided space only. If the space is not enough, you must use the back of the same page only.

ลงชื่อนิสิต.....

วันที่.....

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Part I: Basic Skill 1 & 2

1. (12 points) Find the reduced row echelon of the following matrices.

   a. $\begin{bmatrix} 2 & -3 & 11 \\ 1 & 5 & 7 \\ 1 & 0 & 5 \end{bmatrix}$

   b. $\begin{bmatrix} 7 & 1 & -6 & 0 \\ 6 & 0 & -2 & 4 \\ 2 & 2 & 3 & 3 \end{bmatrix}$

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2. (3 point) Write the system of linear equations defined by the following augmented matrix.

$$\left[\begin{array}{cccccc|c} 1 & -2 & 2 & 0 & 3 & 1 & \\ 1 & -1 & 1 & 3 & 2 & 2 & \\ 0 & 0 & 0 & 1 & 1 & 3 & \end{array}\right]$$

3. (3 point) Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $x = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$.
   a. Show how Ax can be computed using dot product.
   b. Show how Ax can be computed as a linear combination of the columns of A
   c. Find Ax.

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4. (12 points) Find the solution <u>set</u> of the systems of linear equations defined by the following augmented matrices. Compute the augmented matrices in RRE form. Identify homogeneous and particular parts of the solution. State which are pivot and free variables.

   a. $\left[\begin{array}{cccccc|c} 1 & 2 & 2 & 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 & 2 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 3 \end{array}\right]$

   b. $\left[\begin{array}{cccc|c} 0 & 2 & 0 & 4 & 8 \end{array}\right]$

   c. $\left[\begin{array}{ccccccc|c} 1 & 2 & 0 & 3 & 0 & 1 & 4 & 4 \\ 0 & 0 & 1 & 2 & 1 & 0 & 5 & 5 \\ 0 & 0 & 0 & 0 & 1 & 0 & 6 & 6 \\ 0 & 0 & 0 & 0 & 0 & 1 & 7 & 7 \end{array}\right]$

   d. $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{array}\right]$

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5. (1.5,1.5,6 points) Find the determinant of the following matrices.

a. $\begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix}$ b. $\begin{bmatrix} 2 & 4 & 2 \\ 3 & 7 & 6 \\ -1 & 4 & 5 \end{bmatrix}$ c. $\begin{bmatrix} 0 & 2 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 4 & 1 \\ 1 & 0 & 1 & 0 & 5 \\ 0 & 0 & 3 & 1 & 1 \end{bmatrix}$

6. (3 point) Compute $AB + 2CC$ where $A = \begin{bmatrix} 1 & -2 & 1 \\ 1 & -1 & 2 \end{bmatrix}, B = \begin{bmatrix} 4 & 5 \\ 3 & -2 \\ 4 & 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$

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7. (9 points) Find a linear equation describing the plane containing the points (1,2,3), (-1,0,1) and (1,1,-1) in $\mathbb{R}^3$.

8. (3,6 points) Find the inverse of the following matrices.

a. $\begin{bmatrix} a & c \\ b & d \end{bmatrix}$ b. $\begin{bmatrix} 1 & 0 & -1 & -1 \\ 1 & 2 & 0 & -1 \\ 1 & -1 & 0 & 1 \\ 1 & -1 & 1 & 2 \end{bmatrix}$

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Part II (2 points per question): Provide short answer only.

1. Give an example of an **eigenvector** together with its **eigenvalue** for the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix}$

   Eigenvector = _____

   Eigenvalue = _____

2. Fill in the blank of the following matrices so that each of them is invertible.

   a. $\begin{bmatrix} 2 & 4 & __ \\ 0 & 7 & 1 \\ 0 & 7 & __ \end{bmatrix}$

   b. $\begin{bmatrix} 2 & __ & 2 \\ 3 & 0 & 6 \\ __ & __ & 5 \end{bmatrix}$

   c. $\begin{bmatrix} 1 & 2 & 4 & 2 \\ 2 & __ & 8 & 4 \\ 3 & 6 & __ & 6 \\ -1 & -2 & -4 & __ \end{bmatrix}$

3. Find two 2×2 matrices A and B such that both are nonsingular but $C = A + B$ is singular.

   a. A =

   b. B =

4. Consider two vectors in $\boldsymbol{R}^2$, $v_1 = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ and $v_1 = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$. We want to find the middle point $p = \begin{pmatrix} x_m \\ y_m \end{pmatrix}$ such that the length from $v_1$to p is equal to the length from $v_2$to p. Let $f(x_1, y_1, x_2, y_2)$ be the function that compute (x_m, y_m). Write a matrix A that represent f

   A =

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Part III: Provide detailed answer.

1. (4 points) Consider a 2×2 matrix A
   a. Prove or disprove that there exists a matrix B such that $A^T = BA$ for all matrices A

   b. Prove or disprove that there exists a sequence of matrix $B_1, B_2, B_3, \dots, B_k$ such that $A^T = B_1 B_2 B_3 \dots B_k A$ for all matrices A

2. Consider the function $F(x_1, x_2, x_3, x_4) = (x_1 + x_2, x_2 - x_3, x_4, x_3 + x_4)$. Answer the following questions.
   a. (1 points) Write a matrix A that represents F

   b. (2 points) Determine that F is one-to-one and/or onto?

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3. (5 points) Let L_1 be the line defined by $\{\lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} | \lambda \in \mathbb{R}\}$ and let L_2 be the line passing through the points $\begin{pmatrix} 0 \\ -2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$. Find the shortest distance between the two lines.

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4. (5 points) Let F be a linear transformation that maps $\begin{pmatrix}1\\2\\3\end{pmatrix}$ to $\begin{pmatrix}1\\2\end{pmatrix}$, and maps $\begin{pmatrix}0\\1\\1\end{pmatrix}$ to $\begin{pmatrix}3\\4\end{pmatrix}$.
Describe the matrices that represents all possible such transformations.

5. (5 points) Let $A = \begin{bmatrix}1 & 2 & 1 & 1 & 0 & 1\\2 & 3 & 1 & 2 & 1 & 2\\1 & 3 & 2 & 2 & 0 & 3\\2 & 4 & 2 & 3 & 1 & 4\end{bmatrix}$ and let $C = \begin{bmatrix}1 & 0 & -1 & 0 & 1 & -1\\0 & 1 & 1 & 0 & -1 & 0\\0 & 0 & 0 & 1 & 1 & 2\end{bmatrix}$ be the reduced row echelon form of A where the zero-row is removed. Find matrix B such that $A = BC$.